Status	Finished
Started	Tuesday, 27 October 2026, 8:01 AM
Completed	Tuesday, 27 October 2026, 8:31 AM
Duration	30 mins
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓

b. Hash

c. Both (1) & (3)

d. Linear

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓

b. By sorting the list first

c. By dividing the list into halves

d. By using a hash function

Question **3**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Insertion

c. Selection

d. Bubble ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues in the left or right sublist ✓
c. The search stops
d. The search continues from the beginning

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Linear search ✓
c. Interpolation search
d. Binary search

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Searching ✓
c. Mining
d. Finding

Question **8**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is sorted ✓
c. When the list is unsorted
d. When the list is very small

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. At the beginning of the list
c. In the right sublist
d. In the entire list

Question **11**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is large and unsorted
c. When the list is large and sorted
d. When the list is small and unsorted ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O👎
- c. O(log n)
- d. O(n log n)

Question **13**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search starts over from the beginning
- b. The search continues until the end of the list
- c. The search backtracks to find duplicate elements
- d. The search stops immediately ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. The list must be sorted
- b. There must be a mechanism to delete and/or insert elements in the list ✓
- c. There should be direct access to the middle element in any sublist
- d. Number values should only be present

Number values should only be present

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. There must be a mechanism to access middle element directly
- d. Must use a sorted array